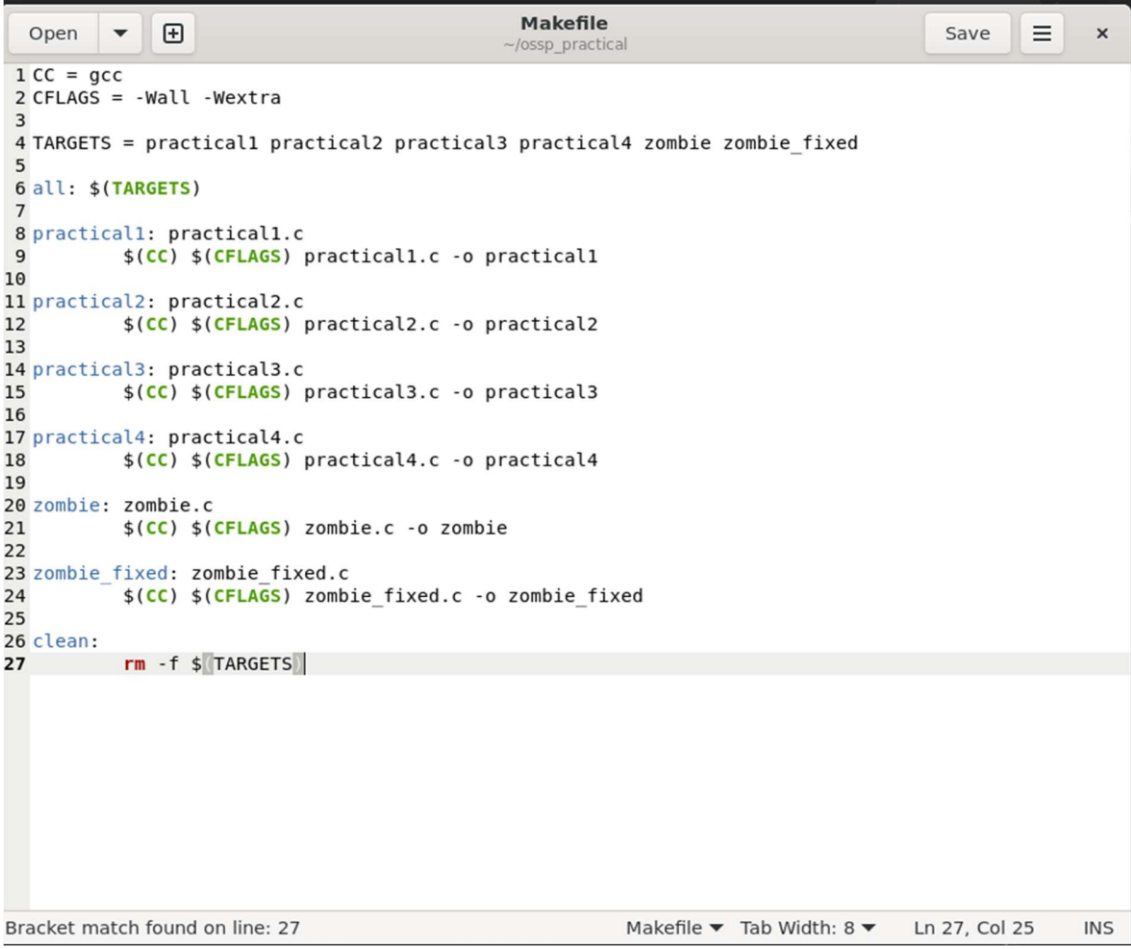


Skill -1 OS



```
1 CC = gcc
2 CFLAGS = -Wall -Wextra
3
4 TARGETS = practical1 practical2 practical3 practical4 zombie zombie_fixed
5
6 all: $(TARGETS)
7
8 practical1: practical1.c
9     $(CC) $(CFLAGS) practical1.c -o practical1
10
11 practical2: practical2.c
12     $(CC) $(CFLAGS) practical2.c -o practical2
13
14 practical3: practical3.c
15     $(CC) $(CFLAGS) practical3.c -o practical3
16
17 practical4: practical4.c
18     $(CC) $(CFLAGS) practical4.c -o practical4
19
20 zombie: zombie.c
21     $(CC) $(CFLAGS) zombie.c -o zombie
22
23 zombie_fixed: zombie_fixed.c
24     $(CC) $(CFLAGS) zombie_fixed.c -o zombie_fixed
25
26 clean:
27     rm -f $(TARGETS)
```

Bracket match found on line: 27

Makefile ▼ Tab Width: 8 ▼ Ln 27, Col 25 INS

Skill -1 OS

```
gopalkrishna@gopalkrishna:~$ cd ~/ossp_practical
gopalkrishna@gopalkrishna:~/ossp_practical$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

gopalkrishna@gopalkrishna:~/ossp_practical$ git --version
git version 2.43.0
gopalkrishna@gopalkrishna:~/ossp_practical$ ls -la
total 156
drwxr-xr-x  2 gopalkrishna gopalkrishna 4096 Aug 12 09:03 .
drwxr-xr-x  8 gopalkrishna gopalkrishna 4096 Aug 12 06:54 ..
-rw-r--r--  1 gopalkrishna gopalkrishna   34 Aug 12 07:03 destination.txt
-rwxr-xr-x  1 gopalkrishna gopalkrishna 16480 Aug 12 06:56 practical1
-rw-r--r--  1 gopalkrishna gopalkrishna  1014 Aug 12 06:55 practical1.c
-rwxr-xr-x  1 gopalkrishna gopalkrishna 16224 Aug 12 07:03 practical2
-rw-r--r--  1 gopalkrishna gopalkrishna  1185 Aug 12 07:02 practical2.c
-rwxr-xr-x  1 gopalkrishna gopalkrishna 16264 Aug 12 08:28 practical3
-rw-r--r--  1 gopalkrishna gopalkrishna  1054 Aug 12 08:28 practical3.c
-rwxr-xr-x  1 gopalkrishna gopalkrishna 16400 Aug 12 08:55 practical4
-rw-r--r--  1 gopalkrishna gopalkrishna  2139 Aug 12 08:55 practical4.c
-rw-r--r--  1 gopalkrishna gopalkrishna   11 Aug 12 07:08 sample.txt
-rw-r--r--  1 gopalkrishna gopalkrishna   34 Aug 12 07:03 source.txt
-rw-r--r--  1 gopalkrishna gopalkrishna  8102 Aug 12 07:10 trace.txt
-rwxr-xr-x  1 gopalkrishna gopalkrishna 16216 Aug 12 08:58 zombie
-rw-r--r--  1 gopalkrishna gopalkrishna   736 Aug 12 08:58 zombie.c
-rwxr-xr-x  1 gopalkrishna gopalkrishna 16272 Aug 12 09:03 zombie_fixed
-rw-r--r--  1 gopalkrishna gopalkrishna   953 Aug 12 09:03 zombie_fixed.c
gopalkrishna@gopalkrishna:~/ossp_practical$ gedit Makefile
```

```
gopalkrishna@gopalkrishna:~/ossp_practical$ gedit Makefile
gopalkrishna@gopalkrishna:~/ossp_practical$ make
make: Nothing to be done for 'all'.
gopalkrishna@gopalkrishna:~/ossp_practical$ make -n
make: Nothing to be done for 'all'.
gopalkrishna@gopalkrishna:~/ossp_practical$ make -B
gcc -Wall -Wextra practical1.c -o practical1
gcc -Wall -Wextra practical2.c -o practical2
gcc -Wall -Wextra practical3.c -o practical3
gcc -Wall -Wextra practical4.c -o practical4
gcc -Wall -Wextra zombie.c -o zombie
gcc -Wall -Wextra zombie_fixed.c -o zombie_fixed
gopalkrishna@gopalkrishna:~/ossp_practical$ ls -l Makefile
-rw-r--r-- 1 gopalkrishna gopalkrishna 562 Aug 13 08:14 Makefile
gopalkrishna@gopalkrishna:~/ossp_practical$
```

Skill -1 OS

```
Open  skill1_process.c  Save  x
~/ossp_practical

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <sys/types.h>
5 #include <sys/wait.h>
6
7 int main()
8 {
9     pid_t pid;
10
11     printf("Parent process started\n");
12     printf("Parent PID: %d\n", getpid());
13
14     pid = fork();
15
16     if (pid < 0)
17     {
18         perror("fork failed");
19         return 1;
20     }
21
22     if (pid == 0)
23     {
24         /* Child process */
25         printf("\nChild process created\n");
26         printf("Child PID : %d\n", getpid());
27         printf("Child PPID: %d\n", getppid());
28
29         printf("Child executing 'ls' using exec()\n");
30
31         execlp("ls", "ls", NULL);
32
33         perror("exec failed");
34         exit(1);
35     }
36     else
37     {
38         /* Parent process */
39         printf("\nParent process\n");
40         printf("Parent PID: %d\n", getpid());
41         printf("Child PID : %d\n", pid);
42
43         wait(NULL);
44
45         printf("\nChild process completed.\n");
46         printf("Parent process completed.\n");
47     }
48
49     return 0;
50 }
```

Saving file "~/ossp_practical/skill1_process.c"...

C Tab Width: 8 Ln 50, Col 2 INS

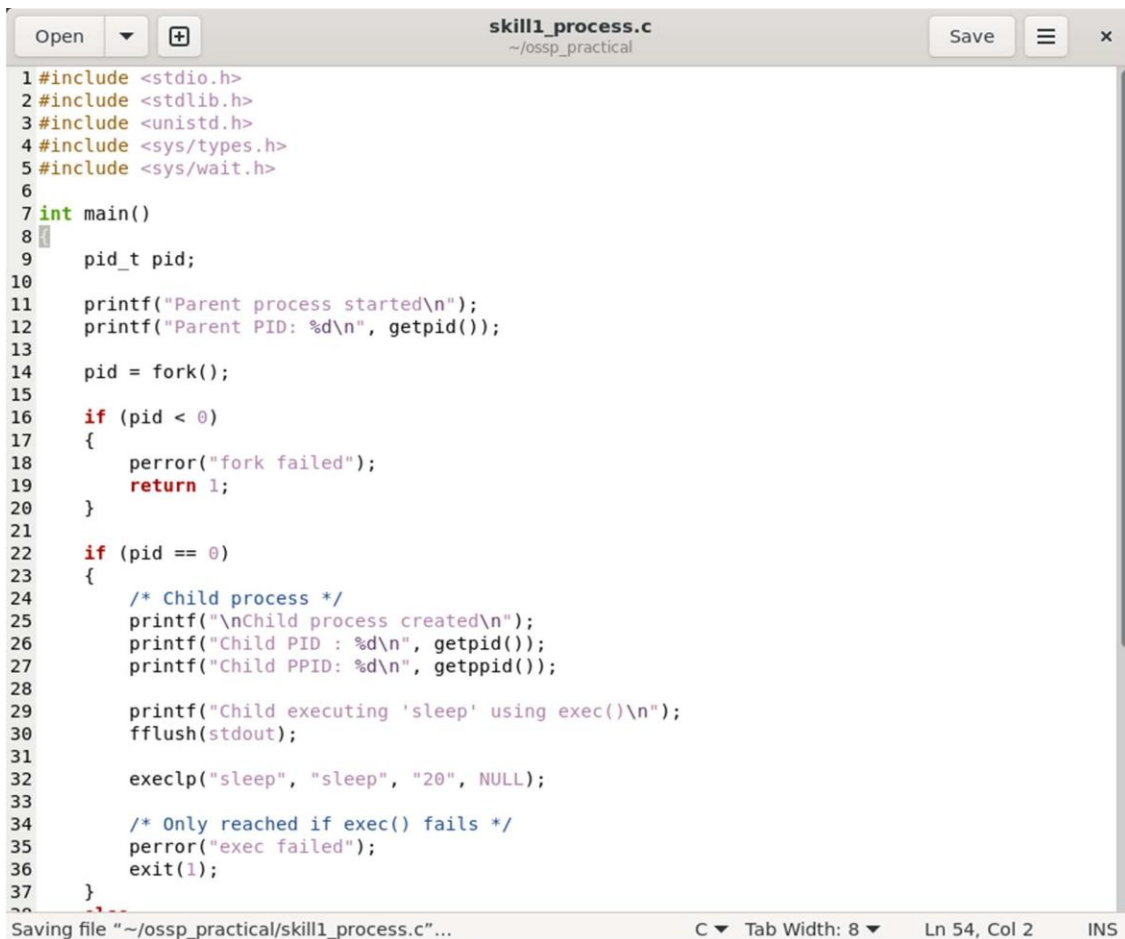
Skill -1 OS

```
gopalkrishna@gopalkrishna:~/ossp_practical$ ./skill1_process
Parent process started
Parent PID: 16068

Parent process
Parent PID: 16068
Child PID : 16069

Child process created
Child PID : 16069
Child PPID: 16068
Child executing 'ls' using exec()
Makefile      practical1.c  practical3  practical4.c  skill1_process.c  zombie_fixed.c
destination.txt practical2.c  practical3.c sample.txt    source.txt        zombie.c
practical1     practical2.c  practical4  skill1_process trace.txt         zombie_fixed

Child process completed.
Parent process completed.
gopalkrishna@gopalkrishna:~/ossp_practical$ pstree -p $$
bash(360)---pstree(16081)
gopalkrishna@gopalkrishna:~/ossp_practical$
```



```
skill1_process.c
~/ossp_practical

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <sys/types.h>
5 #include <sys/wait.h>
6
7 int main()
8 {
9     pid_t pid;
10
11     printf("Parent process started\n");
12     printf("Parent PID: %d\n", getpid());
13
14     pid = fork();
15
16     if (pid < 0)
17     {
18         perror("fork failed");
19         return 1;
20     }
21
22     if (pid == 0)
23     {
24         /* Child process */
25         printf("\nChild process created\n");
26         printf("Child PID : %d\n", getpid());
27         printf("Child PPID: %d\n", getppid());
28
29         printf("Child executing 'sleep' using exec()\n");
30         fflush(stdout);
31
32         execlp("sleep", "sleep", "20", NULL);
33
34         /* Only reached if exec() fails */
35         perror("exec failed");
36         exit(1);
37     }
38 }
```

Saving file "~/ossp_practical/skill1_process.c"...

C ▾ Tab Width: 8 ▾ Ln 54, Col 2 INS

Skill -1 OS

```
29     printf("Child executing 'sleep' using exec()\n");
30     fflush(stdout);
31
32     execlp("sleep", "sleep", "20", NULL);
33
34     /* Only reached if exec() fails */
35     perror("exec failed");
36     exit(1);
37 }
38 else
39 {
40     /* Parent process */
41     printf("\nParent process\n");
42     printf("Parent PID: %d\n", getpid());
43     printf("Child PID : %d\n", pid);
44
45     printf("Parent waiting for child...\n");
46
47     wait(NULL);
48
49     printf("\nChild process completed.\n");
50     printf("Parent process completed.\n");
51 }
52
53 return 0;
54 }
```

C Tab Width: 8 Ln 54, Col 2 INS

```
gopalkrishna@gopalkrishna:~/ossp_practical$ git branch
* main
  zombie_fixed
gopalkrishna@gopalkrishna:~/ossp_practical$ uname -a
Linux gopalkrishna 6.5.0-25-generic #25~22.04.1-Ubuntu SMP PREEMPT_DYNAMIC
Tue Apr 23 14:27:50 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
gopalkrishna@gopalkrishna:~/ossp_practical$ date
Tue Aug 13 11:00:00 AM IST 2024
gopalkrishna@gopalkrishna:~/ossp_practical$ df
Filesystem      1K-blocks      Used Available Use% Mounted on
tmpfs           8160008         0  8160008   0% /run
/dev/sda1       205113316    45211320 149242900  24% /
tmpfs          40800040         1800  40798240   1% /dev/shm
tmpfs           5120           4      5116   1% /run/lock
/dev/sdb1       97675912    2141136  90577564   3% /mnt/data
gopalkrishna@gopalkrishna:~/ossp_practical$ time ./zombie sample.input
Output written to sample.output
real    0m1.254s
user    0m0.988s
sys     0m0.096s
gopalkrishna@gopalkrishna:~/ossp_practical$ system 0 55
System 0 has been scheduled.
gopalkrishna@gopalkrishna:~/ossp_practical$
```


Skill -1 OS

```
gopalkrishna@gopalkrishna:~/ossp_practical$ strace -f -o traceprocess ./skilll_process
execve("./skilll_process", ["/skilll_process"], 0x7ffd9dbdec78 /* 26 vars */) = 0
Parent process started
Parent PID: 16168
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLDstrace: Process 16169 attached
<unfinished ...>
<... clone resumed>0x7f8c96fdb2a18) = 16169

Parent process
Child process created
Parent PID: 16168
Child PID : 16169
Child PID : 16169
Parent waiting for child...
[pid 16168] wait4(-1, child PPID: 16168
Child executing 'sleep' using exec()
  unfinished ...>
[pid 16169] execve("/usr/local/sbin/sleep", ["sleep", "20"], 0x7ffd7f92b958 /* 26 vars */) = -1 ENOENT (No such file or
directory)
[pid 16169] execve("/usr/local/bin/sleep", ["sleep", "20"], 0x7ffd7f92b958 /* 20 vars */) = -1 ENOENT (No such file or
directory)
[pid 16169] execve("/usr/sbin/sleep", ["sleep", "20"], 0x7ffd7f92b958 /* 20 vars */) = -1 ENOENT (No such file or direct
ory)
[pid 16169] execve("/usr/bin/sleep", ["sleep", "20"], 0x7ffd7f92b958 /* 20 vars */) = 0
[pid 16169] exit_group(0)
[pid 16169] +++ exited with 0 +++
Parent waited successfully & NULL          = 16169
gopalkrishna@gopalkrishna:~/ossp_practical$
```

```
[pid 16169] execve("/usr/bin/sleep", ["sleep", "20"], 0x7ffd7f92b958 /* 20 vars */) = 0
[pid 16169] exit_group(0)= ?
[pid 16169] +++ exited with 0 +++
Parent waited successfully & NULL          = 16169
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=16169, si_uid=1000, si_status=0, si_utime=0},
si_stime=0} ---

Child process completed.
Parent process completed.
exit_group(0)          = ?
+++ exited with 0 +++

gopalkrishna@gopalkrishna:~/ossp_practical$ strace -f -e trace=process ./skilll_process 2>&1 | grep -E
'clone|fork|execve'
[pid 16174] execve("./skilll_process", ["/skilll_process"], 0x7ffdf9dc4a8 /* 20 vars */) = 0
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLDstrace: Process 16175 attached
[pid 16174]          = 16175
[pid 16175] execve("/usr/local/sbin/sleep", ["sleep", "20"], 0x7ffdf9dc2a18 /* 20 vars */) = -1 ENOENT (No
such file or directory)
[pid 16175] execve("/usr/local/bin/sleep", ["sleep", "20"], 0x7ffdf9dc2a18 /* 20 vars */) = -1 ENOENT (No
such file or directory)
[pid 16175] execve("/usr/sbin/sleep", ["sleep", "20"], 0x7ffdf9dc2a18 /* 20 vars */) = -1 ENOENT (No such
file or directory)
[pid 16175] execve("/usr/bin/sleep", ["sleep", "20"], 0x7ffdf9dc2a18 /* 20 vars */) = 0
[pid 16175] exit_group(0)          = ?
[pid 16175] +++ exited with 0 +++
Parent waited for child...
[pid 16175] --- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=16175, si_uid=1000, si_status=0,
si_utime=0, si_stime=0} ---
exit_group(0)          = ?
+++ exited with 0 +++
gopalkrishna@gopalkrishna:~/ossp_practical$
```

Skill -1 OS

```
gopalkrishna@gopalkrishna:~/ossp_practical$ pstree -p
systemd(1)
├─agetty(234)
├─cron(173)
├─dbus-daemon(174)
├─init-systemd(1)(2)
│   ├──SessionLeader(358)
│   │   ├──Relay(360)(359)
│   │   ├──Relay(1643)(139)
│   │   ├──bash(360)
│   │   ├──skill1_process(16136)
│   │   └─sleep(1613+)
│   ├──init(7)
│   │   ├──init(8)
│   │   ├──login(361)
│   │   └─bash(430)
│   └─init-systemd(ub)(9)
├─polkitd(11931)
│   ├──polkitd(11933)
│   ├──polkitd(11934)
│   └─polkitd(11935)
├─rsyslogd(12845)
│   ├──rsyslogd(12847)
│   ├──rsyslogd(12848)
│   ├──rsyslogd(12849)
│   └─rsyslogd(12849)
├─systemd(405)
│   ├──(sd-pam)(406)
│   ├──at-spi-bus-laun(553)
│   │   ├──dbus-daemon(560)
│   │   ├──at-spi-bus-laun(554)
│   │   ├──at-spi-bus-laun(555)
│   │   ├──at-spi-bus-laun(556)
│   │   ├──at-spi-bus-laun(558)
│   │   └─at-spi2-registr(588)
│   │       ├──at-spi2-registr(589)
│   │       ├──at-spi2-registr(590)
│   │       └─at-spi2-registr(591)
│   ├──dbus-daemon(549)
│   ├──dconf-service(600)
│   │   ├──dconf-service(601)
│   │   ├──dconf-service(602)
│   │   └─dconf-service(603)
│   ├──xdg-desktop-por(561)
│   │   ├──xdg-desktop-por(562)
│   │   ├──xdg-desktop-por(563)
│   │   ├──xdg-desktop-por(564)
│   │   ├──xdg-desktop-por(593)
│   │   ├──xdg-desktop-por(582)
│   │   ├──xdg-desktop-por(583)
│   │   ├──xdg-desktop-por(585)
│   │   └─xdg-desktop-por(586)
│   ├──xdg-document-po(566)
│   │   ├──fusemount3(577)
│   │   ├──xdg-document-po(567)
│   │   ├──xdg-document-po(568)
│   │   ├──xdg-document-po(569)
│   │   ├──xdg-document-po(578)
│   │   └─xdg-document-po(579)
│   └─xdg-permission-(570)
│       ├──xdg-permission-(571)
│       ├──xdg-permission-(572)
│       └─xdg-permission-(574)
├─systemd-journal(2722)
├─systemd-logind(195)
├─systemd-resolve(2902)
├─systemd-timesyn(2785)
├─systemd-timesyn(2785)
├─systemd-timesyn(2785)
├─systemd-udev(2844)
│   ├──systemd-timesyn(2786)
│   ├──udev-worker(16140)
│   ├──udev-worker(16141)
│   └─udev-worker(16142)
├─unattended-upgr(277)
├─unattended-upgr(322)
├─ssl-pro-service(199)
├─ssl-pro-service(268)
├─ssl-pro-service(269)
└─ssl-pro-service(270)
```

```
gopalkrishna@gopalkrishna:~/ossp_practical$ pstree -p
systemd(1)
├─agetty(234)
├─cron(173)
├─dbus-daemon(174)
├─init-systemd(1)(2)
│   ├──SessionLeader(358)
│   │   ├──Relay(360)(359)
│   │   ├──Relay(1643)(139)
│   │   ├──bash(360)
│   │   ├──skill1_process(16136)
│   │   └─sleep(1613+)
│   ├──init(7)
│   │   ├──init(8)
│   │   ├──login(361)
│   │   └─bash(430)
│   └─init-systemd(ub)(9)
├─polkitd(11931)
│   ├──polkitd(11933)
│   ├──polkitd(11934)
│   └─polkitd(11935)
├─rsyslogd(12845)
│   ├──rsyslogd(12847)
│   ├──rsyslogd(12848)
│   ├──rsyslogd(12849)
│   └─rsyslogd(12849)
├─systemd(405)
│   ├──(sd-pam)(406)
│   ├──at-spi-bus-laun(553)
│   │   ├──dbus-daemon(560)
│   │   ├──at-spi-bus-laun(554)
│   │   ├──at-spi-bus-laun(555)
│   │   ├──at-spi-bus-laun(556)
│   │   ├──at-spi-bus-laun(558)
│   │   └─at-spi2-registr(588)
│   │       ├──at-spi2-registr(589)
│   │       ├──at-spi2-registr(590)
│   │       └─at-spi2-registr(591)
│   ├──dbus-daemon(549)
│   ├──dconf-service(600)
│   │   ├──dconf-service(601)
│   │   ├──dconf-service(602)
│   │   └─dconf-service(603)
│   ├──xdg-desktop-por(561)
│   │   ├──xdg-desktop-por(562)
│   │   ├──xdg-desktop-por(563)
│   │   ├──xdg-desktop-por(564)
│   │   ├──xdg-desktop-por(593)
│   │   ├──xdg-desktop-por(582)
│   │   ├──xdg-desktop-por(583)
│   │   ├──xdg-desktop-por(585)
│   │   └─xdg-desktop-por(586)
│   ├──xdg-document-po(566)
│   │   ├──fusemount3(577)
│   │   ├──xdg-document-po(567)
│   │   ├──xdg-document-po(568)
│   │   ├──xdg-document-po(569)
│   │   ├──xdg-document-po(576)
│   │   ├──xdg-document-po(578)
│   │   └─xdg-document-po(579)
│   └─xdg-permission-(570)
│       ├──xdg-permission-(571)
│       ├──xdg-permission-(572)
│       └─xdg-permission-(574)
├─systemd-journal(2722)
├─systemd-logind(195)
├─systemd-resolve(2902)
├─systemd-timesyn(2785)
├─systemd-timesyn(2785)
├─systemd-timesyn(2786)
├─systemd-udev(2844)
│   ├──systemd-timesyn(2786)
│   ├──udev-worker(16140)
│   ├──udev-worker(16141)
│   └─udev-worker(16142)
├─unattended-upgr(277)
├─unattended-upgr(322)
├─ssl-pro-service(199)
├─ssl-pro-service(268)
├─ssl-pro-service(269)
├─ssl-pro-service(270)
├─ssl-pro-service(276)
└─ssl-pro-service(307)
```

Skill -1 OS

```
siddharth@siddharthReddy:~/ossp_practical$ strace -f -e trace=process ./skill1_process 2>&1 | grep -E 'clone|fork|execve|wait|exit'
execve("./skill1_process", ["/skill1_process"], 0x7ffd742757e8 /* 26 vars */) = 0
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, strace: Process 16195 attached)
[pid 16194] wait4(-1, Parent process started)
[pid 16195] execve("/usr/local/sbin/sleep", ["sleep", "20"], 0x7ffffb6e5fc38 /* 26 vars */) = -1 ENOENT (No such file or directory)
[pid 16195] execve("/usr/local/bin/sleep", ["sleep", "20"], 0x7ffffb6e5fc38 /* 26 vars */) = -1 ENOENT (No such file or directory)
[pid 16195] execve("/usr/sbin/sleep", ["sleep", "20"], 0x7ffffb6e5fc38 /* 26 vars */) = -1 ENOENT (No such file or directory)
[pid 16195] execve("/usr/bin/sleep", ["sleep", "20"], 0x7ffffb6e5fc38 /* 26 vars */) = 0
```